CCSA104 Practical Assignment: Managing Security Policies

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Date	06-11-2026	Score	/ 100

Create these objects in Task 2. The addresses are examples. Use the addressing in your lab topology, and use the name of your own gateway object wherever the tasks say **Gateway**.

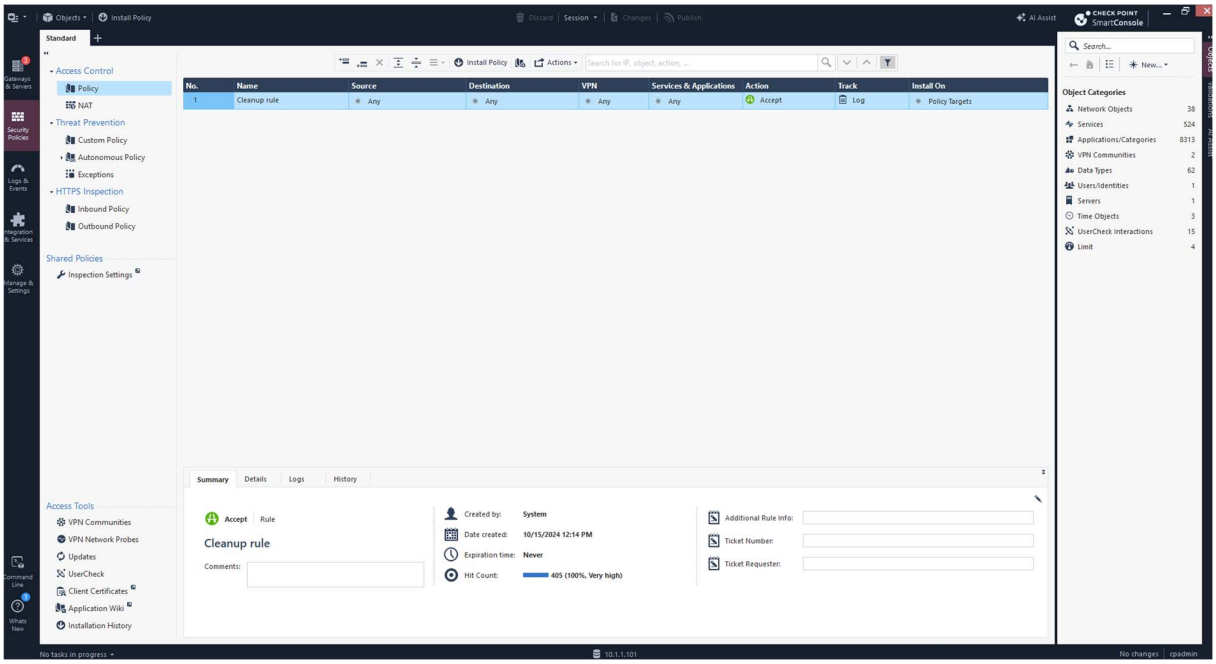
Object name	Type	Example value	Purpose
Internal_LAN	Network	10.1.1.0 / 255.255.255.0	Staff network
DMZ_Net	Network	172.16.10.0 / 255.255.255.0	DMZ network
Admin_PC	Host	10.1.1.101	Administrator workstation
Web_Server	Host	172.16.10.20	Public web server
Mail_Server	Host	172.16.10.25	Mail server
External_DNS	Host	192.9.100.10	External DNS server

Part A: Practical Tasks

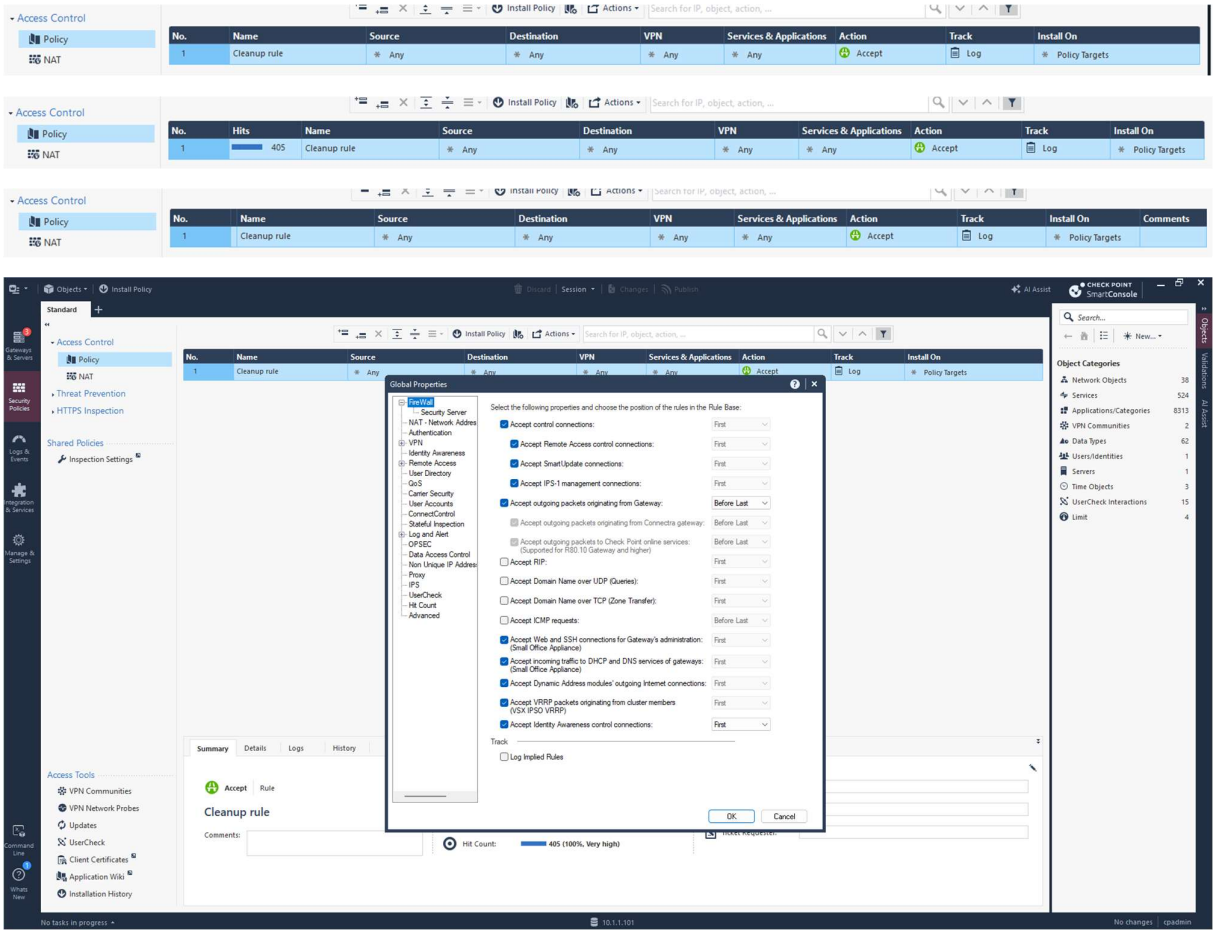
Task 1: Explore the default policy

1. Log in to SmartConsole and open the **Security Policies** view. Record the name of the policy package in use.
2. Find the default **Cleanup rule**. Record its Name, Source, Destination, Services & Applications, Action, Track and Install On values.
3. Right-click a column header in the rulebase. Show the **Hits**, **Track** and **Install On** columns, hide one column, then show it again.
4. From the Applications menu, open **Global Properties** and name three categories of settings you can see. Then open **Manage policies and layers** and list the policy packages that exist.

Evidence:



Policy Package name: Standard



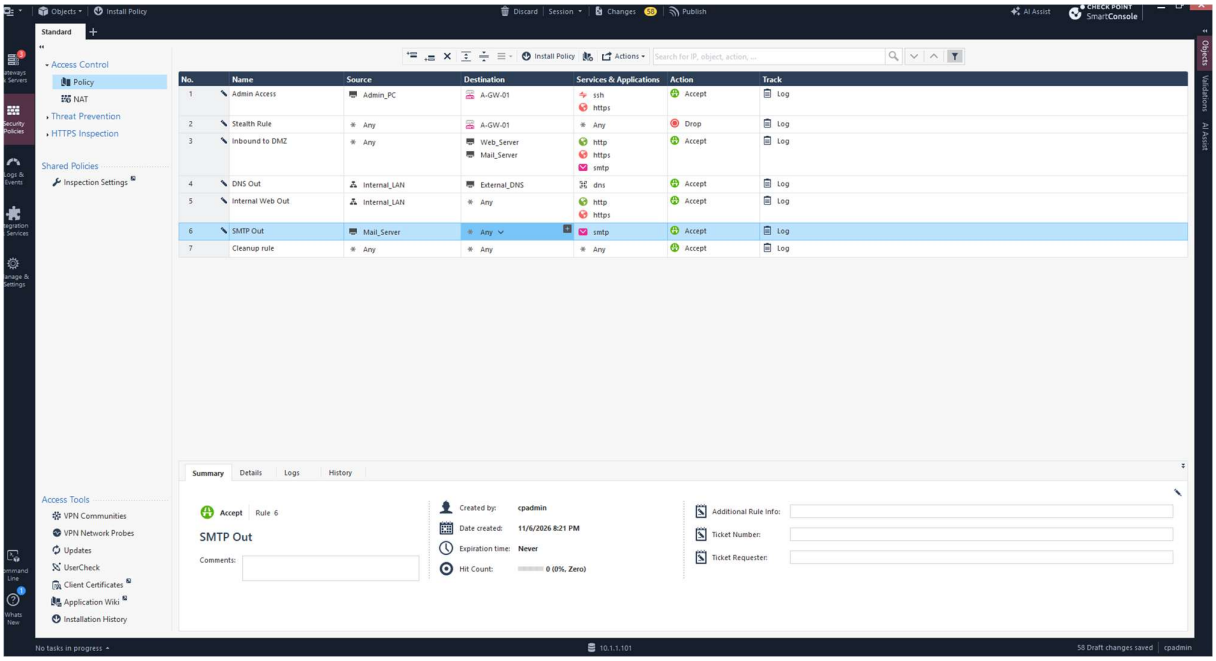
Task 2: Build the Access Control rulebase

Insert new rules above the cleanup rule so that the rulebase matches the table below. SmartConsole numbers rules automatically, so the positions shown are the final positions after Task 5 adds rule 4.

Final position	Name	Source	Destination	Services & Applications	Action	Track
1	Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Inbound to DMZ	Any	Web_Server, Mail_Server	http, https, smtp	Accept	Log
4	Internal to DMZ	*added in Task 5*				
5	DNS Out	Internal_LAN	External_DNS	dns (UDP and TCP)	Accept	Log
6	Internal Web Out	Internal_LAN	Any	http, https	Accept	Log
7	SMTP Out	Mail_Server	Any	smtp	Accept	Log
8	Cleanup Rule	Any	Any	Any	Drop	Log

The Cleanup rule already exists. Do not recreate it. Change its Track setting to **Log** so that dropped connections are recorded, and leave it at the bottom.

Evidence:

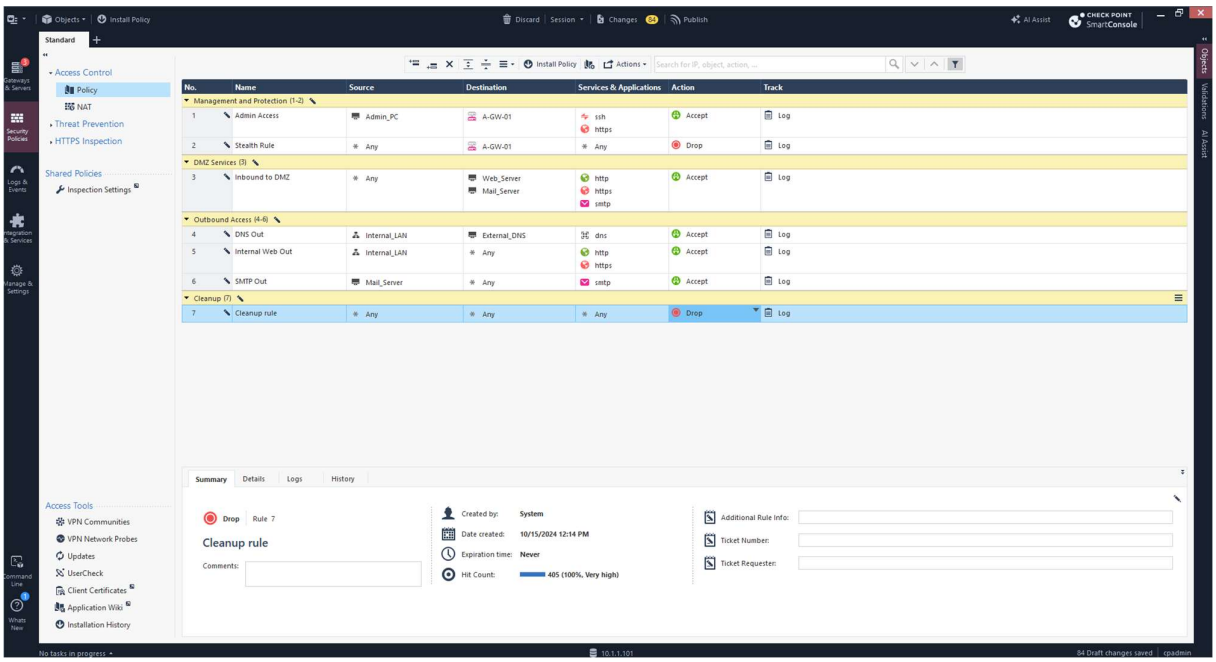


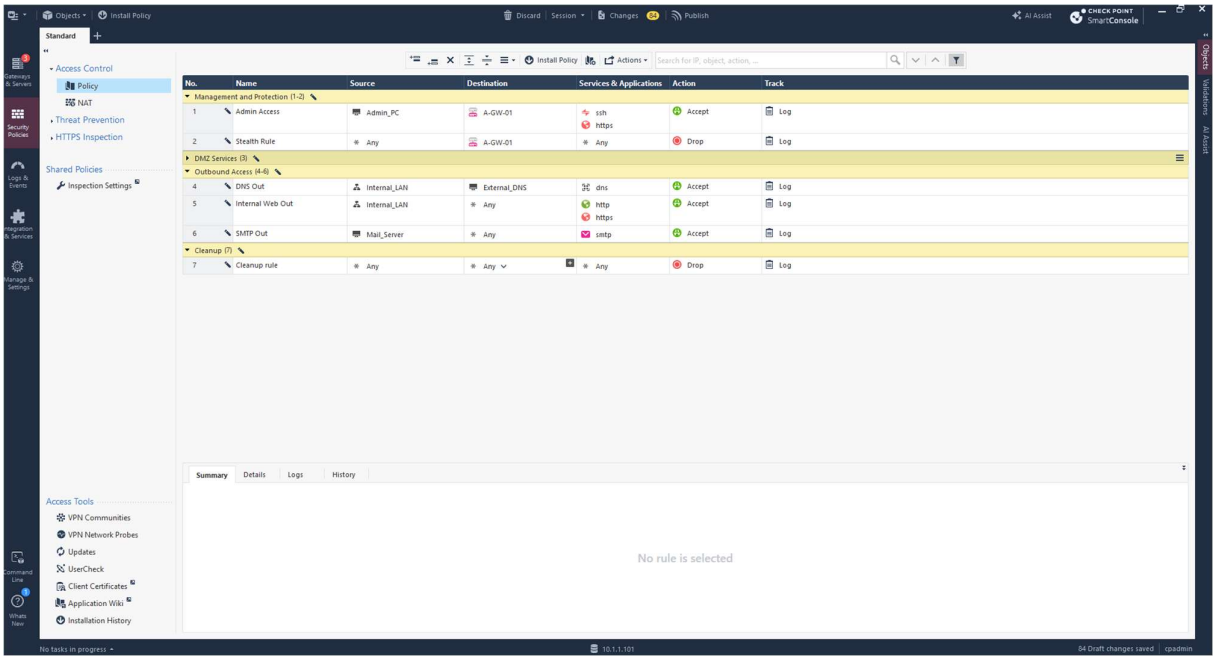
Task 4: Organise the rulebase with sections

Right-click the rulebase, choose **New Section Title**, and select Above or Below to create four sections. Place each rule in the correct section.

Section title	Contains
Management and Protection	Rules 1 and 2
DMZ Services	Rules 3 and 4
Outbound Access	Rules 5, 6 and 7
Cleanup	Rule 8

Evidence:



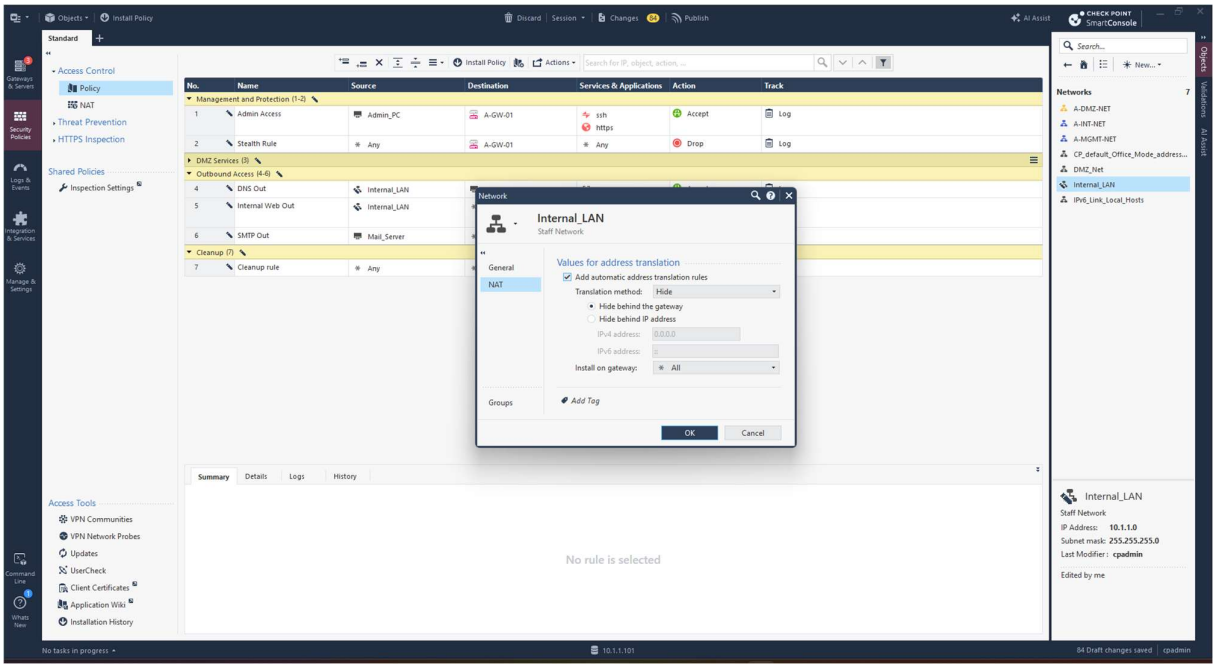


Rulebase sections are purely organizational and do not affect the order in which security rules are enforced; section titles are not sent to the Security Gateway during policy installation

Task 6: Configure Hide NAT

1. Open the **Internal_LAN** object, go to its **NAT** page, enable automatic address translation rules, choose **Hide**, and hide the network behind the gateway.
2. Publish your changes.
3. In the Security Policies view, open the **NAT** rulebase and locate the rule or rules created automatically for Internal_LAN.
4. In your own words, explain why this is an *automatic* NAT rule rather than a manual one, what *many-to-one* translation means, and what the gateway does with the source port.

Evidence:



The screenshot shows the Network Address Translation (NAT) settings for the Internal LAN network object in Check Point SmartConsole.

The Add automatic address translation rules option is enabled, with the translation method set to Hide and Hide behind the gateway selected. This configuration instructs Check Point to automatically generate the necessary Hide NAT rule or rules for the Internal LAN network.

Hide NAT allows multiple internal devices to access external networks using the Security Gateway's translated IP address. The gateway distinguishes individual connections using port address translation (PAT), allowing multiple users to share a single external IP address.

The Install on gateway setting is configured as All, meaning the automatically generated NAT rules are eligible for installation on applicable gateways.

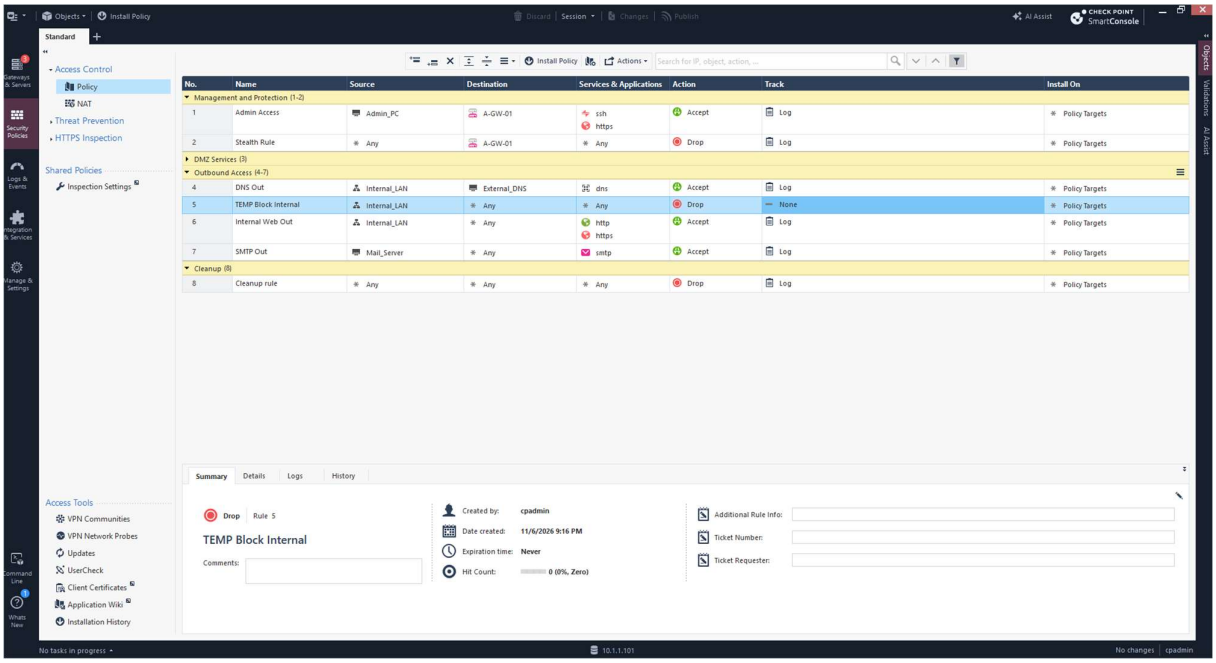
Once the configuration is saved and published, the automatically generated rules can be viewed under Security Policies → Access Control → NAT, typically within the automatic NAT rules section.

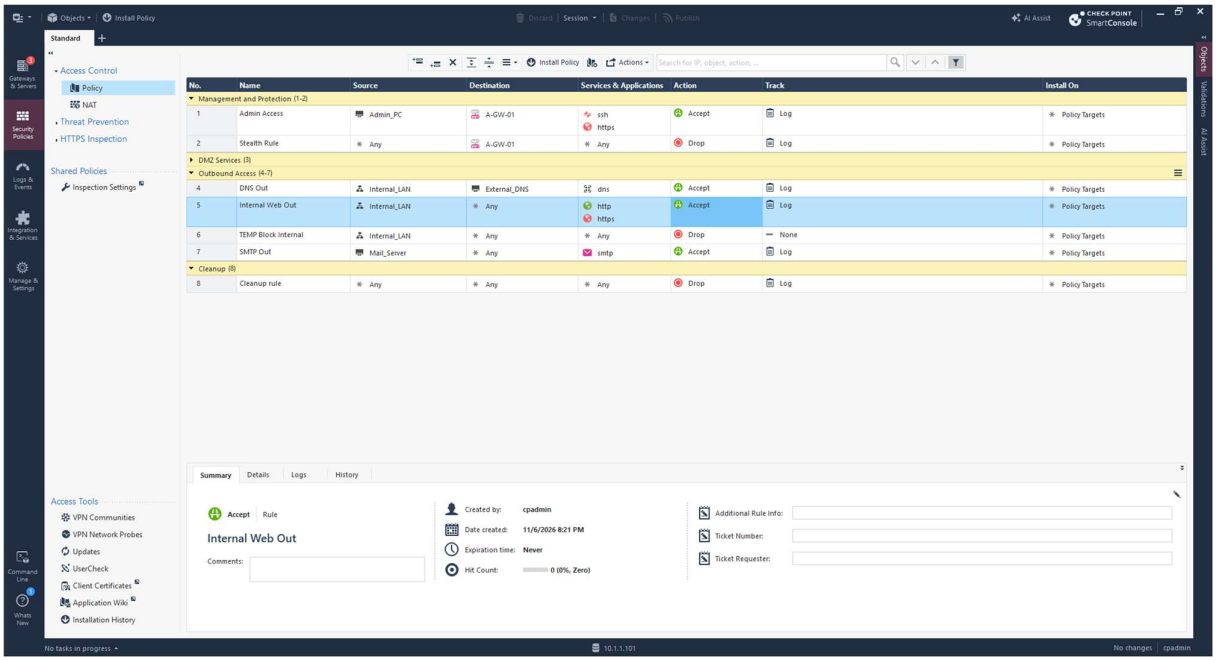
Conclusion: The screenshot confirms that automatic Hide NAT has been configured for Internal LAN. However, it shows the network object's NAT settings rather than the generated NAT rulebase itself. Successful NAT operation must still be verified by installing the policy and examining translated connections in the logs or through appropriate packet inspection

Task 7: Rule order experiment and Installation History (5 marks)

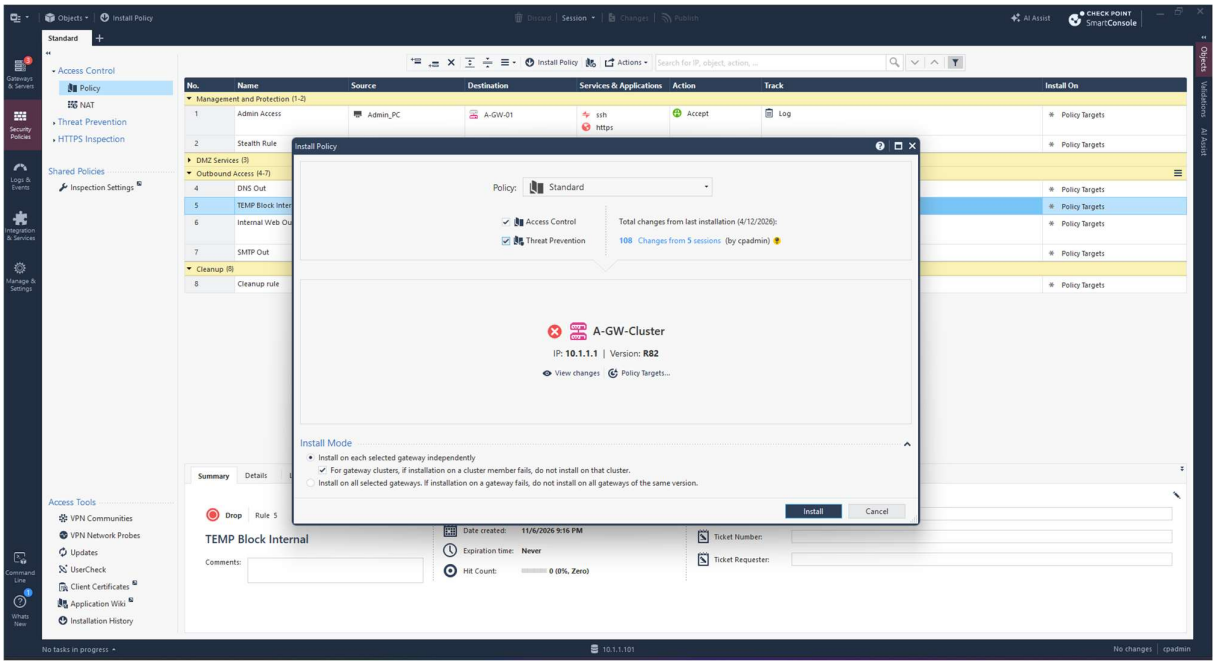
1. Add a temporary rule named **TEMP Block Internal** (Source Internal_LAN, Destination Any, Services Any, Action Drop, Track Log) **above** Internal Web Out. Publish and install. Repeat test T4 and note which rule matches.
2. Move **TEMP Block Internal below** Internal Web Out. Publish and install. Repeat T4 and note which rule matches. Explain the difference.
3. Delete the temporary rule, then publish and install again.
4. In the Security Policies view, open **Access Tools** and then **Installation History**. Identify the installation that contained the temporary rule and open its read-only or audit view. Do not revert unless your instructor says so.

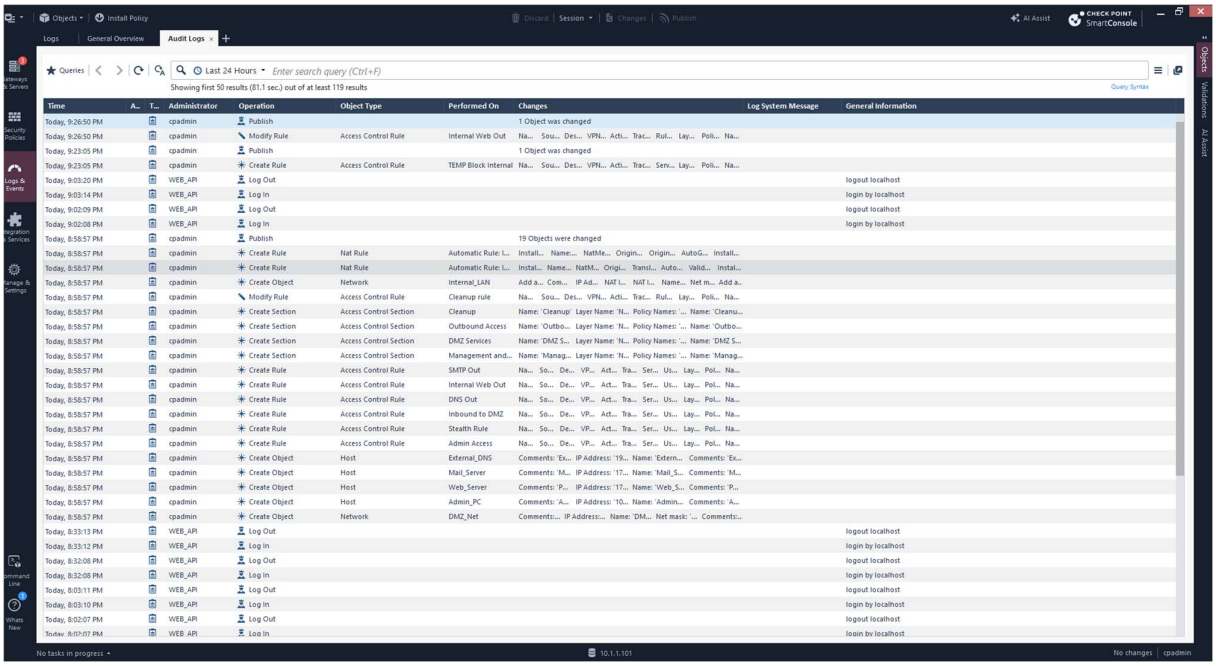
Evidence:





log screenshots for steps 1 and 2, the Installation History screenshot, and a three-sentence explanation of why specific rules belong above general rules.




Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Logs
General Overview
Audit Logs
Queries
Last 24 Hours
Enter search query (Ctrl+F)
Showing first 50 results (81.1 sec.) out of at least 119 results
Query Syntax
Time
A...
T...
Administrator
Operation
Object Type
Performed On
Changes
Log System Message
General Information
Today, 9:26:50 PM cpadmin Publish 1 Object was changed
Today, 9:26:50 PM cpadmin Modify Rule Access Control Rule Internal Web Out
Today, 9:23:05 PM cpadmin Publish 1 Object was changed
Today, 9:23:05 PM cpadmin Create Rule Access Control Rule TEMP Block Internal
Today, 9:03:20 PM WEB_API Log Out logout localhost
Today, 9:03:14 PM WEB_API Log In login by localhost
Today, 9:02:09 PM WEB_API Log Out logout localhost
Today, 9:02:08 PM WEB_API Log In login by localhost
Today, 8:58:57 PM cpadmin Publish 19 Objects were changed
Today, 8:58:57 PM cpadmin Create Rule Nat Rule Automatic Rule: I...
Today, 8:58:57 PM cpadmin Create Rule Nat Rule Automatic Rule: I...
Today, 8:58:57 PM cpadmin Create Object Network Internal_LAN
Today, 8:58:57 PM cpadmin Modify Rule Access Control Rule Cleanup rule
Today, 8:58:57 PM cpadmin Create Section Access Control Section Cleanup
Today, 8:58:57 PM cpadmin Create Section Access Control Section Outbound Access
Today, 8:58:57 PM cpadmin Create Section Access Control Section DMZ Services
Today, 8:58:57 PM cpadmin Create Section Access Control Section Management and...
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule SMTP Out
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule Internal Web Out
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule DNS Out
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule Inbound to DMZ
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule Stealth Rule
Today, 8:58:57 PM cpadmin Create Rule Access Control Rule Admin Access
Today, 8:58:57 PM cpadmin Create Object Host External_DNS
Today, 8:58:57 PM cpadmin Create Object Host Mail_Server
Today, 8:58:57 PM cpadmin Create Object Host Web_Server
Today, 8:58:57 PM cpadmin Create Object Host Admin_PC
Today, 8:58:57 PM cpadmin Create Object Network DMZ_Net
Today, 8:33:13 PM WEB_API Log Out logout localhost
Today, 8:33:12 PM WEB_API Log In login by localhost
Today, 8:32:08 PM WEB_API Log Out logout localhost
Today, 8:32:08 PM WEB_API Log In login by localhost
Today, 8:03:11 PM WEB_API Log Out logout localhost
Today, 8:03:10 PM WEB_API Log In login by localhost
Today, 8:02:07 PM WEB_API Log Out logout localhost
Today, 8:02:07 PM WEB_API Log In login by localhost
No tasks in progress
10.1.1.101
No changes
cpadmin

Part B: Theory Questions

Q1. Where is a Security Policy created, stored and enforced?

Answer: B — Created in SmartConsole, stored on the Security Management Server, enforced by Security Gateways.

Explanation: The security administrator creates and manages policies through SmartConsole. Once published, the policies are stored on the Security Management Server and installed on the Security Gateway, which enforces them by inspecting network traffic.

Q2. Which control model does the Check Point Access Control firewall policy use?

Answer: A — Positive Control Model (whitelist).

Explanation: The Positive Control Model allows only traffic that has been explicitly authorized and blocks everything else. This approach helps prevent unauthorized access by ensuring that only approved network communications are permitted.

Q3. Where should the Stealth rule be placed?

Answer: C — As early as possible, typically immediately after the Management rules.

Explanation: The Stealth rule protects the Security Gateway from unauthorized access. Placing it immediately after the Management rules allows legitimate administrative connections while blocking other attempts to access the gateway itself.

Q4. Which statement about rulebase sections is correct?

Answer: D — Sections are visual divisions only and do not affect enforcement order.

Explanation: Rulebase sections help administrators organize rules into logical groups, such as Management and Protection, DMZ Services, Outbound Access, and Cleanup. They make the policy easier to understand and maintain but do not change rule enforcement order or get installed as security rules on the gateway.

Q5. What type of translation does Hide NAT perform?

Answer: B — Many-to-one.

Explanation: Hide NAT allows multiple internal devices to share a single translated IP address when accessing external networks. The gateway uses port translation to distinguish connections from different internal devices.

Section B2: Short Answers

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Answer:

Explicit rules are created manually by the security administrator in the Access Control rulebase. They define which traffic should be allowed or blocked based on conditions such as source, destination, and service.

Implied rules, on the other hand, are automatically generated by Check Point based on predefined system settings. They support essential functions such as communication between the Security Management Server and Security Gateways.

Implied rules are configured under Global Properties → Firewall → Implied Rules. Depending on their settings, they can be placed First, Before Last, or Last in relation to explicit rules.

The Implicit Cleanup Action determines what happens to traffic that does not match an applicable rule. The traditional firewall's implicit cleanup action is Drop by default, although the effective behavior also depends on the Access Control layer's cleanup configuration.

In my lab, I would use an explicit Cleanup rule configured to Drop and Log to ensure unauthorized traffic is blocked and recorded.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Answer:

Rule order matters because Check Point evaluates rules from top to bottom within an ordered Access Control layer. The first matching rule determines how the traffic is handled within that layer.

For example, in my Orion Logistics rulebase, the Admin Access rule comes before the Stealth rule. This allows authorized administrators to access the Security Gateway before the Stealth rule blocks other connections to it.

If I reversed these rules, the Stealth rule could block legitimate administrative access. Therefore, arranging rules in the correct order is important for both security and network functionality.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organizations publish only during a change window?

Answer:

Saving changes in SmartConsole keeps them within the administrator's current session, while publishing commits those changes to the shared Security Management Server database.

Unpublished changes are not automatically included when a policy is installed. SmartConsole may require the administrator to publish the changes before proceeding with installation.

Many organizations schedule publishing and policy installation during approved change windows to allow proper review, coordination, and testing. This reduces the risk of disrupting business operations and makes it easier to identify and reverse problematic changes.

Q9. Compare Global Properties with Policy Packages. Give one example of what each control and say how you open each in SmartConsole.

Answer:

Global Properties contain system-wide settings that influence Check Point security management and firewall behavior. For example, they control the configuration of Implied Rules.

Policy Packages, on the other hand, contain the security policies and rules that are installed on selected Security Gateways. For example, my Standard policy package contains rules for Admin Access, Stealth protection, DMZ services, outbound access, and Cleanup.

To open Global Properties, I select Menu → Global Properties in SmartConsole.

To view Policy Packages, I select Menu → Manage policies and layers, where I can see the available packages and their associated policy layers.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

Answer:

The four predefined Check Point Security Zones are InternalZone, ExternalZone, DMZZone, and WirelessZone.

Two important rules apply when assigning gateway interfaces to zones.

First, each gateway interface can belong to only one Security Zone at a time.

Second, multiple gateway interfaces can belong to the same Security Zone, provided they serve the same security purpose.

One benefit of using zone-based rules is that they simplify security policy management. Instead of creating separate rules for individual IP addresses, administrators can control traffic based on network zones, making policies easier to maintain as the network grows.

Section B3: Scenario Questions

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

Answer:

Placing an Any-to-Any Accept rule above the Stealth rule creates two major security risks.

First, it exposes the Security Gateway to unauthorized access because the broad Accept rule could allow connections to the gateway that the Stealth rule was intended to block.

Second, it weakens the entire security policy by allowing traffic that should have been restricted by later rules. This defeats the Positive Control Model, which requires that only explicitly authorized traffic should be permitted.

To correct this, I would remove or disable the overly permissive rule and replace it with specific rules that permit only the required sources, destinations, and services. I would also ensure that the Stealth rule is positioned immediately after the necessary Management rules.

Management rules are allowed above the Stealth rule because they permit authorized administrators and management systems to communicate with the Security Gateway.

After making the correction, I would publish the changes, install the policy on the appropriate gateway, and verify the results through SmartConsole logs.

Q12. After Hide NAT is configured for Internal LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

Answer:

1. Conservation of public IP addresses: NAT allows multiple internal devices to share one public IP address, reducing the number of public addresses required by the organisation.

2. Hiding internal IP addresses: NAT replaces internal private IP addresses with a translated address, preventing external systems from directly seeing the original internal addresses. However, NAT does not replace the need for proper firewall security rules.

(b) Describe three checks you would make to find the fault, and explain why each one matters.

Answer:

If staff cannot access the Internet after configuring Hide NAT, I would carry out the following checks.

1. Verify the NAT configuration.

I would open the Internal LAN network object in SmartConsole and confirm that automatic NAT is enabled, the translation method is set to Hide, and Hide behind the gateway is selected. I would also check the NAT rulebase for conflicting translation rules.

This is important because an incorrect NAT configuration can prevent internal devices from communicating properly with external networks.

2. Check the Access Control policy and installation status.

I would verify that the firewall has rules permitting Internal LAN users to access required Internet services, particularly HTTP, HTTPS, and DNS. I would also confirm that the latest published policy was successfully installed on the correct Security Gateway.

This matters because NAT only translates addresses; it does not automatically permit traffic. The Access Control policy must also allow the connection.

3. Verify network connectivity, routing, and logs.

I would check that internal workstations have the correct default gateway and DNS settings. I would also verify that the Security Gateway has a valid default route to the Internet.

Finally, I would examine SmartConsole logs to identify dropped connections or translation problems.

This is important because Internet connectivity depends on correct routing, DNS resolution, NAT, and firewall permissions working together.

By checking these three areas, I can identify the cause of the problem and restore Internet access without weakening the organisation's security policy.